

Task 4: Data Visualization



1. Introduction

The purpose of this task is to create visual representations of the cleaned e-commerce dataset. Data visualization helps communicate patterns and insights more clearly than raw numbers alone. The charts in this report show product order counts, product revenue, order status distribution, payment methods, referral sources, and coupon code usage.

2. Visualizations Created

Visualization	Chart Type	Main Purpose
Number of Orders for Each Product	Horizontal bar chart	Compare order quantity by product
Total Revenue Generated by Each Product	Column chart	Compare revenue by product
Status of Orders	Pie chart	Show order status distribution
Distribution of Payment Methods	Doughnut chart	Show payment method usage
Referral Source	Horizontal bar chart	Compare customer referral sources
Distribution of Coupon Codes Used	Pie chart	Show coupon usage

Table 1: Visualizations Created Table

3. Number of Orders for Each Product



Figure 1: Number of Orders for Each Product

Insight: This chart shows the number of orders for each product. Printer had the highest number of orders with 181 orders, followed closely by Tablet and Chair. Phone had the lowest number of orders. This suggests that Printer, Tablet, and Chair were the most frequently purchased products in the dataset.

4. Total Revenue Generated by Each Product

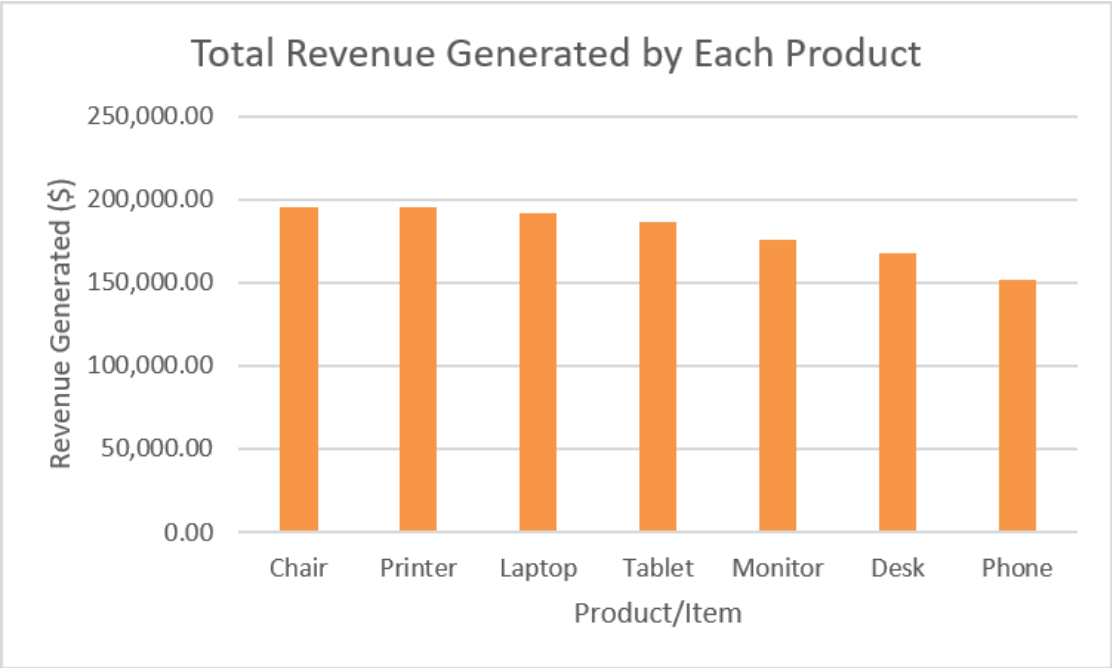


Figure 2: Total Revenue Generated by Each Product

Insight: This chart compares the total revenue generated by each product. Chair generated the highest total revenue, followed very closely by Printer. Although Printer had the most orders, Chair produced slightly higher revenue overall. Phone generated the lowest total revenue.

5. Status of Orders

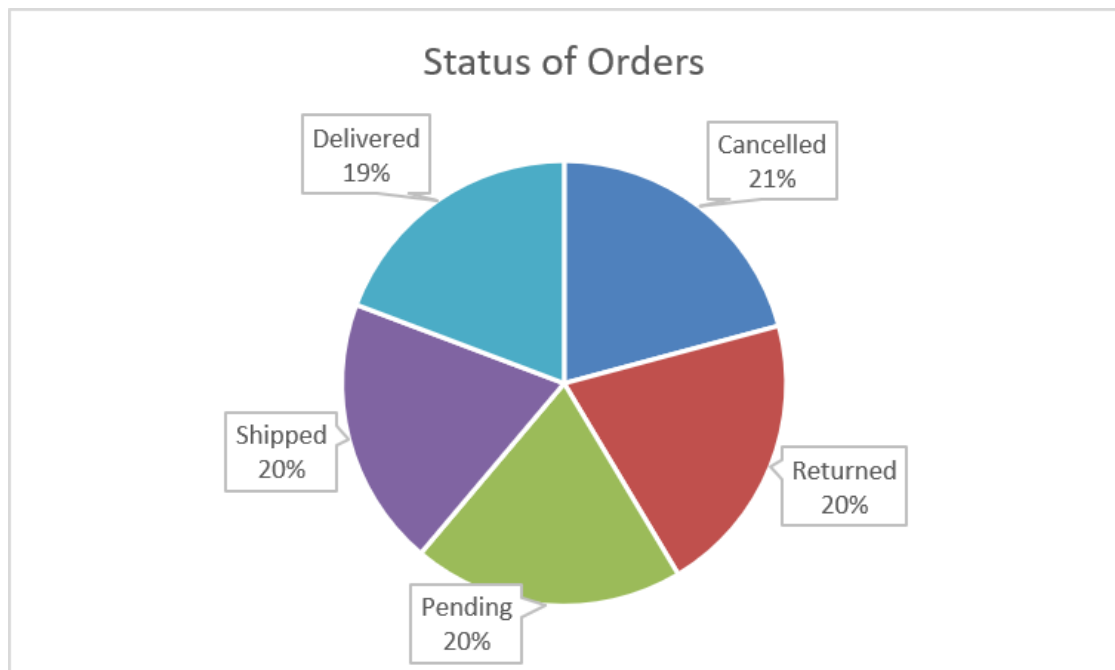


Figure 3: Pie Chart Indicating Order Status

Insight: This pie chart shows the distribution of order statuses. Cancelled orders make up the largest share at 21%, while Returned, Pending, and Shipped orders are each around 20%. Delivered orders are the lowest at 19%. The high percentage of cancelled and returned orders may be an important business concern.

6. Distribution of Payment Methods

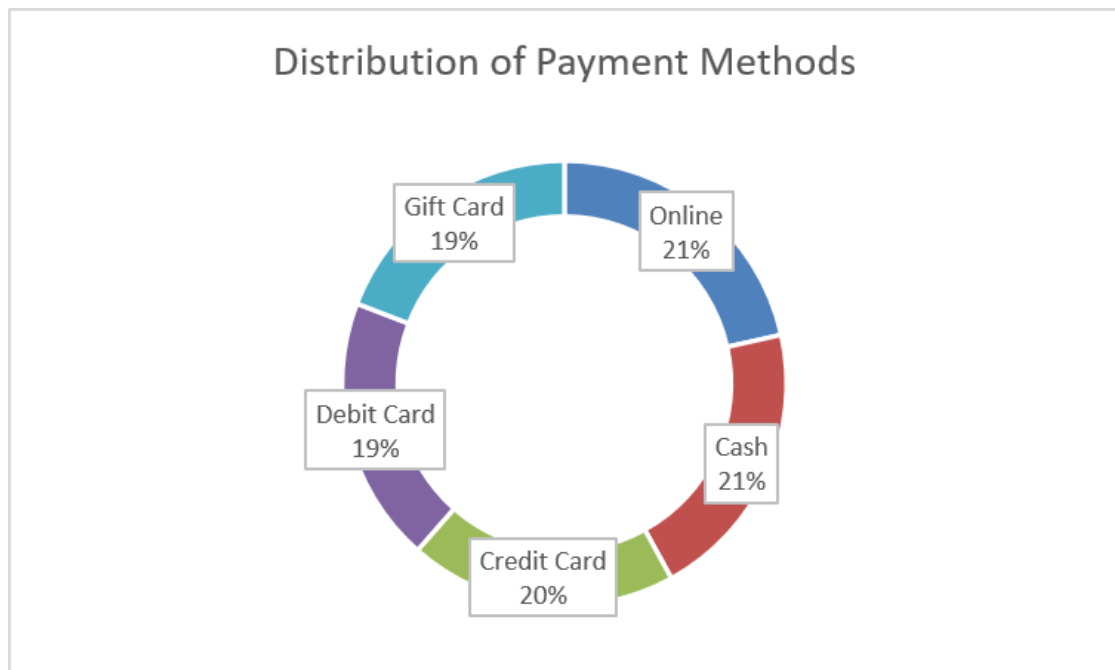


Figure 4: Chart Indicating Distribution of Payment Methods

Insight: This doughnut chart shows how customers paid for their orders. Online and Cash payments each account for about 21% of orders, while Credit Card, Debit Card, and Gift Card payments are close behind. This shows that customers used a variety of payment methods, with no single method dominating strongly.

7. Referral Source

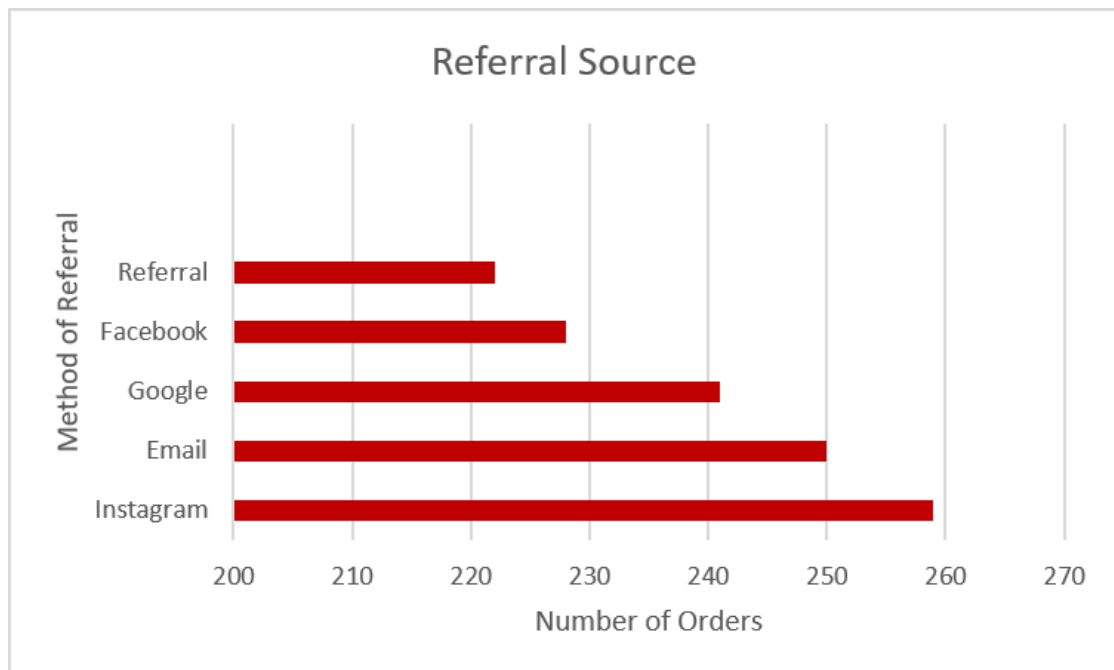


Figure 5: Bar Chart Indicating the Source That Brought the Most Referrals

Insight: This chart shows where customers came from before placing an order. Instagram was the top referral source, followed by Email and Google. Referral and Facebook had lower order counts. This suggests that Instagram and Email may be important channels for attracting customers.

8. Distribution of Coupon Codes Used

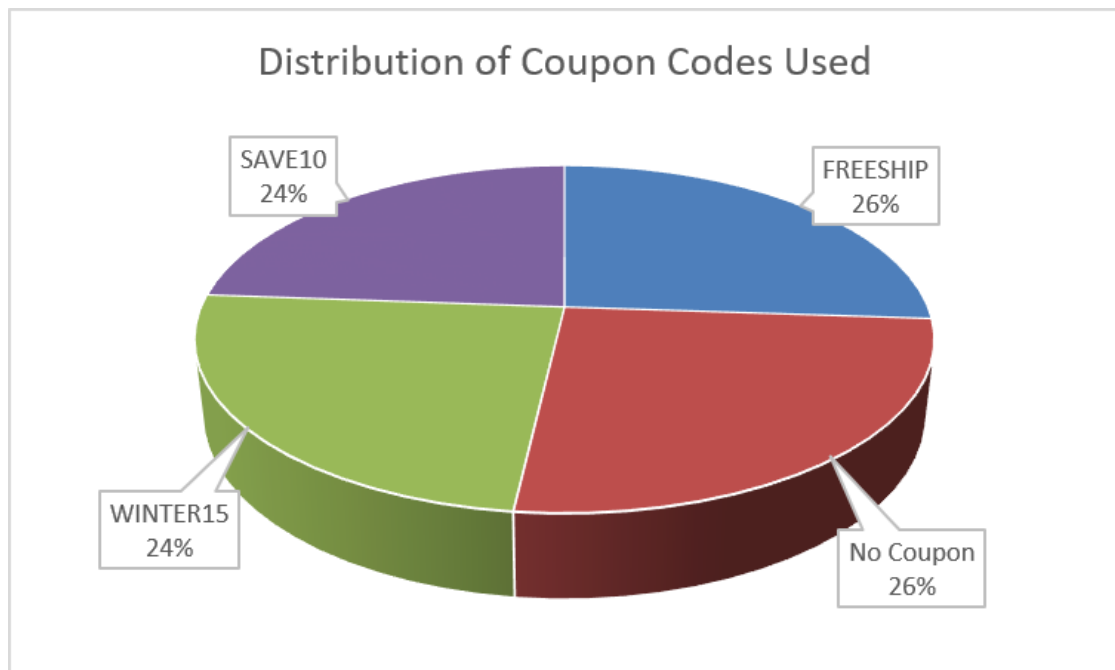


Figure 6: Pie Chart Showing the Most Used Coupon Codes

Insight: This pie chart shows coupon code usage. FREESHIP and No Coupon each account for about 26% of orders, while WINTER15 and SAVE10 each account for about 24%. This shows that coupons were commonly used, but many customers also placed orders without using any coupon.

9. Task 4 Summary

In this task, visualizations were created to communicate important findings from the cleaned e-commerce dataset. The charts showed that Printer had the highest number of orders, while Chair generated the highest total revenue. The order status chart showed that Cancelled and Returned orders were common, which may be a concern for the business. The payment method chart showed that customers used several payment options evenly. The referral source chart showed that Instagram and Email were strong sources of customer traffic. The coupon code chart showed that FREESHIP was commonly used, while many customers also placed orders without a coupon. Overall, these visualizations help explain sales performance, customer behaviour, and possible business concerns in a clear and understandable way.